



Cortical EEG slowing in resting-state Parkinson's disease

Progress update: hypothesis, model, and next steps

PD EEG group · 22.05.2026

Hypothesis (refined)

WHAT WE PREDICT

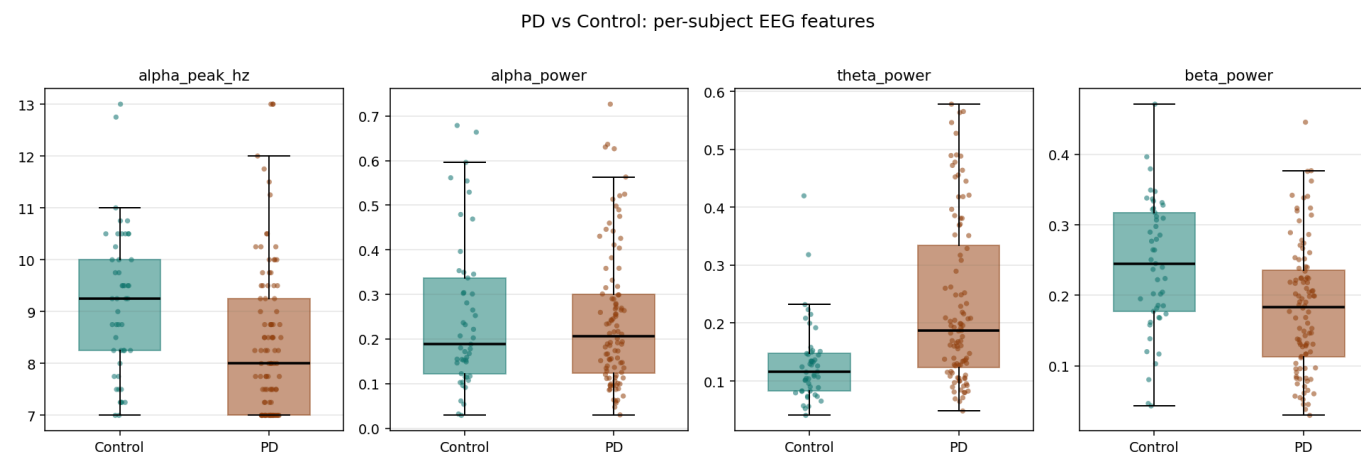
Compared to age-matched healthy controls, Parkinson's disease patients at rest show:

- slowing of the alpha peak frequency (8–13 Hz)
- increased theta-band power (4–8 Hz)
- decreased cortical beta-band power (13–30 Hz)

REFINEMENT BASED ON DATA

Originally formulated as "*decreased alpha power*". Group comparison on 149 subjects showed that alpha power is **essentially unchanged** between groups, while the alpha **peak frequency drops by ~1 Hz in PD** (median 9.25 → 8.00 Hz, $p < 0.001$).

The wording was refined to "*alpha peak slowing*" to reflect the actual change in the data.



Model: Wilson-Cowan network

NETWORK

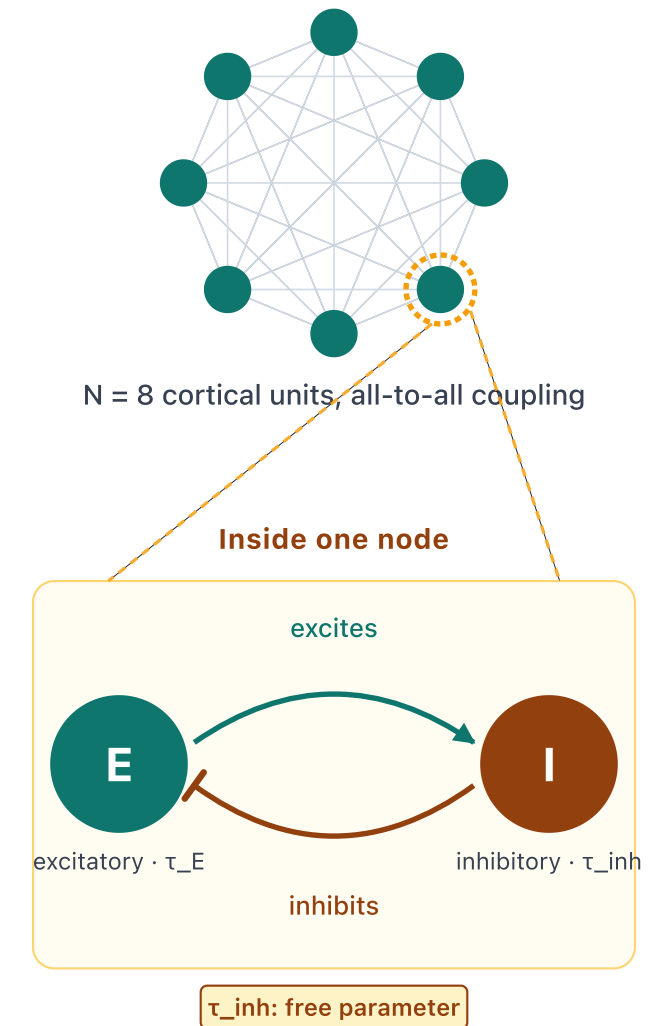
- **8 cortical nodes** connected in a network
- Each node: one excitatory (E) + one inhibitory (I) population
- E and I interact to produce an alpha-range oscillation

FREE PARAMETER

- τ_{inh} : inhibitory response time
- Larger τ_{inh} $\rightarrow$ slower oscillation
- This is the knob we turn to simulate the alpha slowing

SIMPLIFICATIONS

- Identical parameters across all 8 nodes
- All-to-all coupling between nodes
- No noise



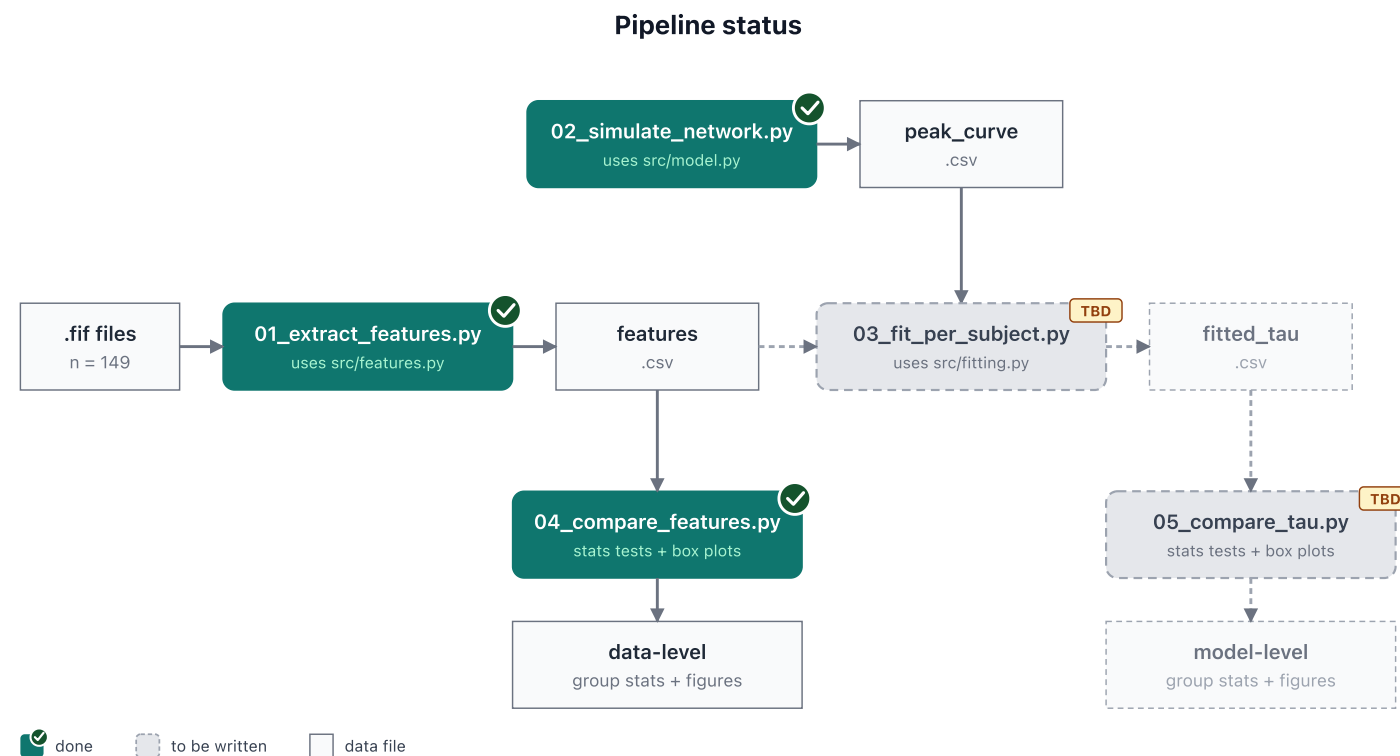
Status and next steps

DONE

- **Feature extraction** on 149 participants
- **Group comparison** on features: all 3 predicted directions significant ($p < 0.001$)
- **Wilson-Cowan network** built and validated (single-node and $N=8$)
- **τ_{inh} sweep**: lookup table built (11 $\rightarrow$ 32 ms)

REMAINING

- **Per-subject fitting** via curve inversion
- **Model-level group test** on fitted τ_{inh}
- State the final " $+\Delta\tau_{inh}$ " result



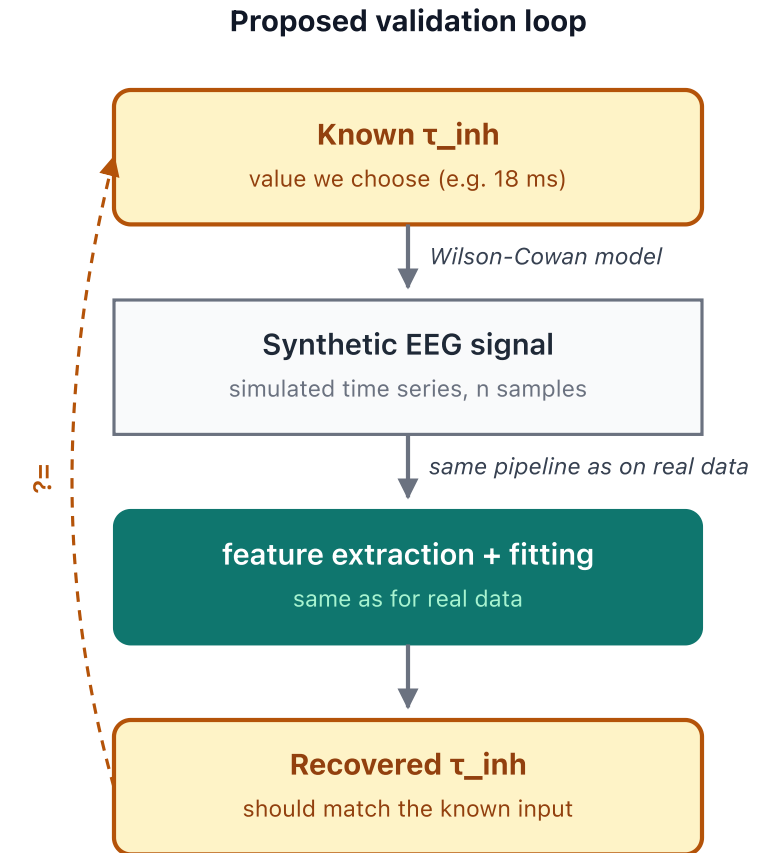
Current challenge

A TEST WE SHOULD RUN (RIGHT)

1. Pick a τ_{inh} value ourselves (e.g., 18 ms)
2. Use the model to generate a synthetic EEG signal at that τ_{inh}
3. Run our fitting procedure on it: does it give back 18 ms?

WHY THIS MATTERS

- We have not yet done this test for our fitting procedure
- Without it, we cannot tell whether PD vs Control differences in fitted τ_{inh} reflect a real effect or errors made by the procedure



If recovered $\approx$ known:
the fitting procedure is sound,
and the differences we find on real data
are likely genuine rather than artifacts.